

Dynamic Mass of Photons and Its Implications for Nonlocality

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Laboratory estimate of the correlation delay: $\Delta t_{\text{corr}} \sim 10^{-21} \text{ s} \rightarrow \lesssim 5 \times 10^{-47} \text{ s}$.

Abstract

This updated work examines the implications of assigning a dynamic, frequency-dependent effective mass to photons within the comprehensive framework of the T0 model, building upon the complete field-theoretic derivation and natural units system where $\hbar = c = \alpha_{\text{EM}} = \beta_{\text{T}} = 1$. The theory establishes the fundamental relationship $T(x, t) = \frac{1}{\max(m, \omega)}$ with dimension $[E^{-1}]$, providing a unified treatment of massive particles and photons through the three fundamental field geometries. The dynamic photon mass $m_\gamma = \omega$ introduces energy-dependent nonlocality effects, with testable predictions. All formulations maintain strict dimensional consistency with the fixed T0 parameters $\beta = 2Gm/r$, $\xi = 2\sqrt{G} \cdot m$, and the cosmic screening factor $\xi_{\text{eff}} = \xi/2$ for infinite fields.

.1 Introduction: T0 Model Foundation for Photon Dynamics

This updated analysis builds upon the comprehensive T0 model framework established in the field-theoretic derivation, incorporating the complete geometric foundations and natural units system. The dynamic effective mass concept for photons emerges naturally from the T0 model's fundamental time-mass duality principle.

Fundamental T0 Model Framework

The T0 model is based on the intrinsic time field definition:

$$T(x, t) = \frac{1}{\max(m(\vec{x}, t), \omega)} \quad (1)$$

Dimensional verification: $[T(x, t)] = [1/E] = [E^{-1}]$ in natural units ✓
This field satisfies the fundamental field equation:

$$\nabla^2 m(\vec{x}, t) = 4\pi G \rho(\vec{x}, t) \cdot m(\vec{x}, t) \quad (2)$$

From this foundation emerge the key parameters:

T0 Model Parameters for Photon Analysis

$$\beta = \frac{2Gm}{r} \quad [1] \text{ (dimensionless)} \quad (3)$$

$$\xi = 2\sqrt{G} \cdot m \quad [1] \text{ (dimensionless)} \quad (4)$$

$$\beta_T = 1 \quad [1] \text{ (natural units)} \quad (5)$$

$$\alpha_{EM} = 1 \quad [1] \text{ (natural units)} \quad (6)$$

Photon Integration in Time-Mass Duality

For photons, the T0 model assigns an effective mass:

$$m_\gamma = \omega \quad (7)$$

Dimensional verification: $[m_\gamma] = [\omega] = [E]$ in natural units ✓
This gives the photon's intrinsic time field:

$$T(x, t)_\gamma = \frac{1}{\omega} \quad (8)$$

Praktische Vereinfachung

Vereinfachung: Da alle Messungen in unserem endlichen, beobachtbaren Universum lokal erfolgen, wird nur die **lokalisierte Feldgeometrie** verwendet:

$\xi = 2\sqrt{G} \cdot m$ und $\beta = \frac{2Gm}{r}$ für alle Anwendungen.

Der kosmische Abschirmfaktor $\xi_{\text{eff}} = \xi/2$ entfällt.

Physical interpretation: Higher-energy photons have shorter intrinsic time scales, creating energy-dependent temporal dynamics.

.2 Energy-Dependent Nonlocality and Quantum Correlations

Entangled Photon Systems

For entangled photons with energies ω_1 and ω_2 , the time field difference is:

$$\Delta T_\gamma = \left| \frac{1}{\omega_1} - \frac{1}{\omega_2} \right| \quad (9)$$

Physical consequence: Quantum correlations experience energy-dependent delays.

Modified Bell Inequality

The energy-dependent time fields lead to a modified Bell inequality:

$$|E(a, b) - E(a, c)| + |E(a', b) + E(a', c)| \leq 2 + \epsilon(\omega_1, \omega_2) \quad (10)$$

where:

$$\epsilon(\omega_1, \omega_2) = \alpha_{\text{corr}} \left| \frac{1}{\omega_1} - \frac{1}{\omega_2} \right| \frac{2G\langle m \rangle}{r} \quad (11)$$

with α_{corr} being a correlation coupling constant and $\langle m \rangle$ the average mass in the experimental setup.

.3 Experimental Predictions and Tests

High-Precision Quantum Optics Tests

Energy-Dependent Bell Tests

Predicted time delay between entangled photons:

$$\Delta t_{\text{corr}} = \frac{G\langle m \rangle}{r} \left| \frac{1}{\omega_1} - \frac{1}{\omega_2} \right| \quad (12)$$

For laboratory conditions with $\langle m \rangle \sim 10^{-3}$ kg, $r \sim 10$ m, and $\omega_1, \omega_2 \sim 1$ eV:

$$\Delta t_{\text{corr}} \lesssim 5 \times 10^{-47} \text{ s} \quad (13)$$

(Corrected on 30 Sept. 2026: previously $\Delta t_{\text{corr}} \sim 10^{-21} \text{ s} - G\langle m \rangle/(rc^2) = 6.674 \times 10^{-11} \cdot 10^{-3}/(10 \cdot c^2) = 7.4 \times 10^{-32}$, $|1/\omega_1 - 1/\omega_2| \leq \hbar/(1 \text{ eV}) = 6.6 \times 10^{-16} \text{ s}$, product $\leq 4.9 \times 10^{-47} \text{ s}$; smaller still for $\omega_1 \approx \omega_2$.)

.4 Dimensional Consistency Verification

Equation	Left Side	Right Side	Status
Photon effective mass	$[m_\gamma] = [E]$	$[\omega] = [E]$	✓
Photon time field	$[T_\gamma] = [E^{-1}]$	$[1/\omega] = [E^{-1}]$	✓
Energy loss rate	$[d\omega/dr] = [E^2]$	$[g_T \omega^2 2G/r^2] = [E^2]$	✓
Time field difference	$[\Delta T_\gamma] = [E^{-1}]$	$[1/\omega_1 - 1/\omega_2] = [E^{-1}]$	✓
Bell correction	$[\epsilon] = [1]$	$[\alpha_{\text{corr}} \Delta T_\gamma \beta] = [1]$	✓

Table 1: Dimensional consistency verification for photon dynamics in T0 model